Water Resources, Dams, Over-Exploitation & Case Study

1️⃣ Introduction to Water Resources

Meaning

Water resources are **natural sources of water that are useful for humans and ecosystems**.

Even though **71% of Earth is covered with water**, only **2.5% is freshwater** and **less than 1% is easily available for human use**.

Types of Water Resources

Surface Water

Water found in rivers, lakes and reservoirs.

Example: **Ganges River**

Groundwater

Water stored underground in **aquifers** and obtained through wells and borewells.

Many cities depend heavily on groundwater.

Rainwater

Rainfall is the **main natural source that refills rivers, lakes and groundwater**.

Rainwater harvesting helps store this water for later use.

Importance of Water Resources

Water resources are essential for:

- Drinking and domestic use
- Agriculture and irrigation
- Industrial activities
- Hydroelectric power generation

- Maintaining ecosystems and biodiversity

Without proper water resources, human development is not possible.

🏞️ 2️⃣ Dams and Their Uses

What is a Dam?

A **dam is a barrier built across a river to store and control water flow**.

The stored water forms a **reservoir** which can be used for different purposes throughout the year.

Example: **Bhakra Nangal Dam**

Uses of Dams

Irrigation

Dams store water and supply it to agricultural lands.

This helps farmers grow crops even during dry seasons.

Hydroelectric Power

Water stored in dams is released through turbines to generate electricity.

Hydropower produces about **16% of global electricity**.

Flood Control

Dams store excess rainwater and release it slowly.

This reduces flood damage in downstream areas.

Drinking Water Supply

Many cities depend on dams for drinking water.

Reservoirs supply water to urban and rural areas.

Recreation and Tourism

Reservoirs created by dams support tourism activities such as boating and fishing.

Advantages of Dams

- Reliable water supply
- Electricity generation
- Agricultural development
- Flood protection

Disadvantages of Dams

- Displacement of people
- Loss of forests and wildlife habitats
- Environmental disturbance
- Sedimentation in reservoirs

🌍 3️⃣ Important Dams That Helped People

Hirakud Dam

The **Hirakud Dam** in **Odisha** was built to control floods in the Mahanadi River.

Before the dam was constructed, the region experienced severe floods frequently.

Today the dam helps regulate river flow and provides irrigation and electricity.

Tehri Dam

The **Tehri Dam** in Uttarakhand helps control floods and generates hydroelectric power.

It also supplies drinking water to major cities including **Delhi**.

Three Gorges Dam

The **Three Gorges Dam** in China on the **Yangtze River** is the largest hydroelectric dam in the world.

It produces about **22,500 MW of electricity** and helps reduce flood risks.

💧 4️⃣ Over-Exploitation of Water Resources

Meaning

Over-exploitation means **using water resources faster than they can naturally replenish**.

This leads to depletion of water resources.

Causes

Major causes include:

- Rapid population growth
- Increased agricultural irrigation
- Industrial expansion
- Excessive groundwater pumping
- Urbanization

Agriculture alone uses about **70% of global freshwater**.

Effects

Over-exploitation can cause:

- Falling groundwater levels
- Drying rivers and lakes
- Land subsidence
- Water scarcity
- Ecological imbalance

For example, groundwater depletion has become a serious issue in **Chennai**.

🌱 5️⃣ Solutions for Sustainable Water Use

To protect water resources, sustainable water management is necessary.

Important solutions include:

Rainwater Harvesting

Collecting and storing rainwater for later use.

Efficient Irrigation

Methods like **drip irrigation** can save **30–50% water** compared to traditional irrigation.

Wastewater Recycling

Treating and reusing wastewater for agriculture or industry.

Water Conservation

Reducing unnecessary water usage in homes, industries and agriculture.

🎯 Conclusion

Water resources are essential for human survival and development.

Dams provide irrigation, electricity and flood control, but they must be managed responsibly.

Over-exploitation of water resources can lead to serious environmental problems.

Therefore, **sustainable water management is necessary to ensure water availability for future generations**.